



CASE HISTORY | WATER & WASTEWATER

Hyderabad's Oldest CETP Modernised for Zero Liquid Discharge

6 MLD Reverse Osmosis & ZLD Upgrade · Jeedimetla Effluent Treatment Limited, Telangana

SITE INFORMATION

- LOCATION:**
Jeedimetla, Hyderabad, Telangana, India
- INDUSTRY:**
Pharmaceutical & Chemical (API)
- APPLICATION:**
CETP Wastewater Recovery, RO + UF, ZLD
- CAPACITY:**
6 MLD permeate
- CLIENT:**
Jeedimetla Effluent Treatment Limited (JETL)
- TDS reduced to <200 PPM
 - 95% system recovery sustained
 - 40+ member industries served
 - One of Hyderabad's oldest CETPs
 - Zero unplanned maintenance stops

TECHNOLOGIES USED

- ▶ Ultrafiltration (UF) Pre-treatment
- ▶ 5-Stage Low Fouling RO
- ▶ Spiral Wound Polyamide RO Elements
- ▶ Zero Liquid Discharge (ZLD)
- ▶ Clarification & Dual Media Filtration
- ▶ PLC-controlled Automation

DELIVERY MODEL

Engineered, supplied, and commissioned by Permionics following a competitive evaluation against four suppliers. Treated water is returned to member industries for process reuse.

OPERATIONS

Continuous CETP operation serving the Jeedimetla industrial estate. Zero unplanned maintenance stops since commissioning, with a flawless compliance record with the State Pollution Control Board.

<200 PPM TDS · 95% Recovery · 6 MLD Capacity · Zero Unplanned Downtime

INTRODUCTION

The Jeedimetla industrial area in Hyderabad is home to one of the region's oldest Common Effluent Treatment Plants (CETPs), serving a cluster of pharmaceutical, API, and chemical manufacturing units operated under Jeedimetla Effluent Treatment Limited (JETL). The plant receives concentrated, highly variable effluent from its member industries and is mandated to recover and reuse treated water in line with Zero Liquid Discharge (ZLD) norms set by the State Pollution Control Board.

JETL approached Permionics to upgrade the CETP's pre-treatment and reverse osmosis blocks, addressing chronic membrane fouling, frequent cleaning cycles, and water recovery shortfalls under the plant's aggressive pharmaceutical-grade effluent load. Permionics deployed a 6 MLD wastewater recovery RO system, engineered for high recovery and low fouling under sustained chemical stress.

THE CHALLENGE

The Problem. The Jeedimetla CETP receives highly concentrated, variable-quality effluent from its member API and chemical units. Aggressive organic and chemical loads caused rapid membrane fouling, forcing frequent cleaning cycles, shortened membrane life, and inconsistent water recovery — all while the plant was required to meet strict ZLD compliance targets set by the State Pollution Control Board.

Existing Limitations. Conventional treatment infrastructure at the estate struggled to cope with the scale and severity of the effluent. Biological treatment stages alone could not reliably handle the high-strength chemical load, and the existing RO system was vulnerable to fouling and scaling, limiting overall plant reliability.

THE TECHNICAL SOLUTION

Why Permionics. Permionics was selected following an extensive evaluation against four competing suppliers, on the strength of its proficiency in engineering and supplying integrated, high-recovery membrane systems. As a fully vertically integrated membrane manufacturer, Permionics formulated customised, low-fouling membrane chemistries specifically suited to withstand the aggressive pharmaceutical effluent at Jeedimetla.

System Design. Permionics deployed a 5-stage low fouling RO system supported by Ultrafiltration (UF) pre-treatment, with clarification and dual media filtration upstream to protect the membranes from suspended solids and organic fouling. The system was engineered to sustain high recovery while resisting the scaling and fouling tendencies of the concentrated CETP effluent.

Post-treatment. The reject stream is routed to downstream Zero Liquid Discharge (ZLD) processing, ensuring no untreated effluent leaves the facility and supporting JETL's compliance with State Pollution Control Board mandates.



The Jeedimetla CETP treatment facility, where clarification and biological treatment stages feed into the Permionics-engineered RO and ZLD system.

Treated Water Quality (Achieved)

Parameter	Unit	Value
Total Dissolved Solids	ppm	<200
System Recovery	%	95%
Unplanned Downtime	days/year	0
Quality-related Rejections	—	Nil

Detailed raw feed-water values (TDS, SDI, TOC, hardness) were not available from the source documents provided and are not included here.



The Permionics low fouling RO skid installed at the Jeedimetla CETP, engineered to sustain high recovery under aggressive pharmaceutical effluent loads.

PERFORMANCE STABILITY

- ▶ 95% water recovery, sustained continuously
- ▶ Zero unplanned maintenance stops since commissioning
- ▶ Flawless compliance record with the State Pollution Control Board
- ▶ Zero product quality rejections attributable to water quality

VALUE DELIVERED

- ▶ TDS reduced to <200 PPM
- ▶ 95% recovery from a historically difficult, high-fouling effluent stream
- ▶ Reduced cleaning frequency and extended membrane life
- ▶ Treated water returned to member industries for process reuse
- ▶ ZLD-aligned operation, eliminating untreated discharge

ENVIRONMENTAL IMPACT

No untreated industrial effluent discharged from the CETP. Purified water returned to member industries for process reuse, supporting ZLD compliance for Hyderabad's oldest API/chemical industrial cluster.

FOR MORE INFORMATION

Permionics Membranes Pvt. Ltd.
www.permionics.com

Water & Wastewater Division
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OPERATIONAL RESULTS

<200 PPM TREATED WATER TDS	95% WATER RECOVERY SUSTAINED CONTINUOUSLY	6 MLD WASTEWATER RECOVERY RO & ZLD PROJECT	0 UNPLANNED DOWNTIME DAYS/YEAR
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Before & After: Visual & Operational Comparison

Parameter	Before Permionics	After Permionics
Treated Water TDS	Elevated, non-compliant	<200 PPM
Water Recovery	Inconsistent	95%, sustained
Unplanned Downtime	Recurring stoppages	0 days since commissioning
Compliance Record	At risk	Flawless with SPCB
Quality-related Rejections	Present	Nil since commissioning

Granular before/after figures for membrane replacement frequency, CIP chemical cost, and specific OPEX (₹/m³) were not available in the source documents provided and have been omitted rather than estimated.



Visual comparison of effluent at different treatment stages at Jeedimetla — from raw, heavily coloured feed (left) through partially treated water (centre) to clear, transparent permeate (right).

FIGURE 1: VISUAL TRANSFORMATION

The progression from raw CETP effluent to fully treated permeate illustrates the scale of the contamination challenge at one of Hyderabad's oldest industrial clusters. Raw effluent drawn from member pharmaceutical and chemical units carried high colour, organic load, and dissolved solids. The Permionics-engineered RO and ZLD system reduces this to a transparent, low-TDS permeate suitable for reuse.

This performance, sustained at 95% recovery with zero unplanned downtime, has supported JETL's ongoing compliance with State Pollution Control Board mandates and eliminated quality-related production rejections among member industries.

CONCLUSIONS

The Jeedimetla CETP upgrade demonstrates how customised, low-fouling membrane chemistry can restore reliability to one of the most demanding effluent treatment challenges in the industry — concentrated, variable pharmaceutical and chemical wastewater at an ageing CETP. By combining Ultrafiltration pre-treatment with a 5-stage low fouling RO system and ZLD-aligned post-treatment, Permionics enabled JETL to sustain 95% water recovery with zero unplanned downtime and a flawless State Pollution Control Board compliance record. Permionics has to date recycled over 10 million m³ of wastewater globally, helping 10,000+ industries achieve their sustainability goals. The Jeedimetla project reinforces that even the oldest, most challenging industrial effluent infrastructure can be modernised into a dependable, ZLD-aligned water recovery asset.